

CSE 2215: Data Structure & Algorithms - I

Application of Stack

Infix Postfix Expression

□ **Infix expression:** The expression of the form $a \text{ op } b$. When an operator is in-between every pair of operands.

□ **Example:** $5 + 8$

□ **Postfix expression:** The expression of the form $a \text{ b op}$. When an operator is followed for every pair of operands.

□ **Example:** $5 \text{ 8 } +$

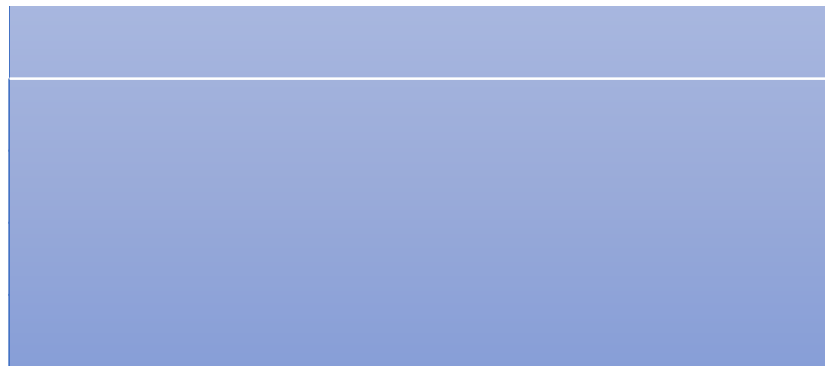
□ **Why postfix representation of the expression?**

□ Infix expressions are readable and solvable by humans. We can easily distinguish the **precedence of operators**, **associativity of operators**, and also know the **uses of parenthesis**.

□ The computer cannot differentiate the operators and parenthesis easily, that's why postfix conversion is needed.

Mathematical Calculations

- The way we usually write expressions is known as infix notation
- What does $3 + 2 * 6$ equal? Ans: 15
- What does $3 * 2 + 6$ equal? Ans: 12
- The precedence of operators affects the order of operations.
- A mathematical expression cannot simply be evaluated left to right.
- Postfix expression does not require any precedence rules.



Algorithm: Convert Infix to Postfix

1. Scan the infix expression from left to right.
2. If the scanned character is an operand, output it.
3. Else [scanned character is an operator]
 - a) if stack is empty or contains a left parenthesis '(' on top, push the incoming operator onto the stack.
 - b) if the incoming symbol is '(', push it onto the stack.
 - c) if the incoming symbol is ')', pop the stack and print the operators until left parenthesis is found.
 - d) if the incoming symbol has higher precedence than the top of the stack, push it onto the stack.
 - e) if the incoming symbol has lower precedence than the top of the stack, pop and print the top. Then test the incoming operator against the new top of the stack.
 - f) if the incoming symbol has equal precedence with the top of the stack, use associativity rule.
 - i) if associativity is L to R, then pop and print the top. Then push the incoming operator.
 - ii) if associativity is R to L, then push the incoming operator.
4. At the end of the expression, pop and print all operators of stack.

Example 1

- Convert the following expressions from infix to postfix:
 - $3 + 2 * 4$

Example 2

- Convert the following expressions from infix to postfix:
 - a) $2 \wedge 3 \wedge 4 + 5 * 7$

Practice

- Convert the following expressions from infix to postfix:

a) $2 \wedge 3 \wedge 4 + 5 * 7$

Ans: $2\ 3\ 4\ \wedge\ \wedge\ 5\ 7\ *\ +$

b) $9 + 2 - 5 * 6 / 3 + 2 \wedge 3 / 4$

Ans: $9\ 2\ +\ 5\ 6\ *\ 3\ /\ -\ 2\ 3\ \wedge\ 4\ /\ +$

c) $1 + 3 \wedge 2 + (5 * 6 - 4) * 7$

Ans: $1\ 3\ 2\ \wedge\ +\ 5\ 6\ *\ 4\ -\ 7\ *\ +$

Evaluation of Postfix Expression

- for each character in postfix expression do
 - if it is an operand then push it onto the stack
 - else if it is an operator (say, op) then
 - pop the stack for the right hand operand (say, b)
 - pop the stack for the left hand operand (say, a)
 - apply the operator to the two operands ($result = a \ op \ b$)
 - push the $result$ onto the stack
 - when the expression has been exhausted the result is the top (and only element) of the stack

Example 1

- What does the following postfix expression evaluate to?

6 3 2 + *

Example 2

- What does the following postfix expression evaluate to?

2 3 4 ^ ^ 5 7 * +